

■ EXAMPLE 2.13

1. Suppose each unit of an electrical instrument is manufactured with a small random bias. Is the noise of the instrumentation ergodic? If we measure the noise of one instrument then we measure its bias, which is equal to its mean. However, if we measure the noise of another instrument it might have a different mean because it has a different bias. In other words, we cannot obtain the mean of the stochastic process by simply investigating one instrument (i.e., one realization of the stochastic process). Therefore, the stochastic process is not ergodic.
2. Suppose each unit of an electrical instrument is manufactured identically, each with zero-mean stationary Gaussian noise. Is the noise ergodic? In this case we could measure the mean of the process by measuring the noise of many separate instruments at one instant of time, or by measuring the noise of one instrument over an extended period of time. Either experiment would correctly inform us that the mean of the stochastic process is zero. We could find the statistics of the stochastic process using all the instruments at a single time, or using a single instrument at many different times. Therefore, the stochastic process is ergodic.

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The definitions of correlation and covariance can be extended to two stochastic processes $X(t)$ and $Y(t)$. The cross correlation of $X(t)$ and $Y(t)$ is defined as

$$R_{XY}(t_1, t_2) = E[X(t_1)Y^T(t_2)] \quad (2.90)$$

Two random processes $X(t)$ and $Y(t)$ are said to be uncorrelated if $R_{XY}(t_1, t_2) = E[X(t_1)]E[Y^T(t_2)]$ for all t_1 and t_2 . The cross covariance of $X(t)$ and $Y(t)$ is defined as

$$C_{XY}(t_1, t_2) = E\{[X(t_1) - \bar{X}(t_1)][Y(t_2) - \bar{Y}(t_2)]^T\} \quad (2.91)$$

2.6 WHITE NOISE AND COLORED NOISE

If the RV $X(t_1)$ is independent from the RV $X(t_2)$ for all $t_1 \neq t_2$ then $X(t)$ is called white noise. Otherwise, $X(t)$ is called colored noise.

The whiteness or color content of a stochastic process can be characterized by its power spectrum. The power spectrum $S_X(\omega)$ of a wide-sense stationary stochastic process $X(t)$ is defined as the Fourier transform of the autocorrelation. The autocorrelation is the inverse Fourier transform of the power spectrum.

$$\begin{aligned} S_X(\omega) &= \int_{-\infty}^{\infty} R_X(\tau) e^{-j\omega\tau} d\tau \\ R_X(\tau) &= \frac{1}{2\pi} \int_{-\infty}^{\infty} S_X(\omega) e^{j\omega\tau} d\omega \end{aligned} \quad (2.92)$$

These equations are called the Wiener–Khinchine relations after Norbert Wiener and Aleksandr Khinchin. Note that some authors put the term $1/2\pi$ on the right side of the $S_X(\omega)$ definition, in which case the $1/2\pi$ term on the right side of the

$R_X(\tau)$ definition disappears. The power spectrum is sometimes referred to as the power density spectrum, the power spectral density, or the power density. The power of a wide-sense stationary stochastic process is defined as

$$P_X = \frac{1}{2\pi} \int_{-\infty}^{\infty} S_X(\omega) d\omega \quad (2.93)$$

The cross power spectrum of two wide-sense stationary stochastic processes $X(t)$ and $Y(t)$ is the Fourier transform of the cross correlation:

$$\begin{aligned} S_{XY}(\omega) &= \int_{-\infty}^{\infty} R_{XY}(\tau) e^{-j\omega\tau} d\tau \\ R_{XY}(\tau) &= \frac{1}{2\pi} \int_{-\infty}^{\infty} S_{XY}(\omega) e^{j\omega\tau} d\omega \end{aligned} \quad (2.94)$$

Similar definitions hold for discrete-time random processes. The power spectrum of a discrete-time random process is defined as

$$\begin{aligned} S_X(\omega) &= \sum_{k=-\infty}^{\infty} R_X(k) e^{-j\omega k} \quad \omega \in [-\pi, \pi] \\ R_X(k) &= \frac{1}{2\pi} \int_{-\pi}^{\pi} S_X(\omega) e^{jk\omega} d\omega \end{aligned} \quad (2.95)$$

A discrete-time stochastic process $X(t)$ is called white noise if

$$\begin{aligned} R_X(k) &= \begin{cases} \sigma^2 & \text{if } k = 0 \\ 0 & \text{if } k \neq 0 \end{cases} \\ &= \sigma^2 \delta_k \end{aligned} \quad (2.96)$$

where δ_k is the Kronecker delta function, defined as

$$\delta_k = \begin{cases} 1 & \text{if } k = 0 \\ 0 & \text{if } k \neq 0 \end{cases} \quad (2.97)$$

The definition of discrete-time white noise shows that it does not have any correlation with itself except at the present time. If $X(k)$ is a discrete-time white noise process, then the RV $X(n)$ is uncorrelated with $X(m)$ unless $n = m$. This shows that the power of a discrete-time white noise process is equal at all frequencies:

$$S_X(\omega) = R_X(0) \quad \text{for all } \omega \in [-\pi, \pi] \quad (2.98)$$

For a continuous-time random process, white noise is defined similarly. White noise has equal power at all frequencies (like white light):

$$S_X(\omega) = R_X(0) \quad \text{for all } \omega \quad (2.99)$$

Substituting this expression for $S_X(\omega)$ into Equation (2.92), we see that for continuous-time white noise

$$R_X(\tau) = R_X(0) \delta(\tau) \quad (2.100)$$

where $\delta(\tau)$ is the continuous-time impulse function. That is, $\delta(\tau)$ is a function that is zero everywhere except at $\tau = 0$; it has a width of 0, a height of ∞ , and an area of 1. Continuous-time white noise is not something that occurs in the real world because it has infinite power, as seen by comparing Equations (2.93) and (2.99). Nevertheless, many continuous-time processes approximate white noise and are useful in mathematical analyses of signals and systems.